VENNY WANJIRU KANGETHE

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Junior Full-Stack Developer

Junior Full-Stack Developer with hands-on experience building scalable web applications using JavaScript, React, Node.js, Next.js, Express, and MongoDB. Strong background in systems support, network operations and improving system uptime. Skilled at designing responsive UIs, building robust RESTful APIs, and applying clean code practices to solve real-world problems. Passionate about delivering user-centered solutions, learning emerging technologies, and contributing to collaborative engineering teams.

KEY HIGHLIGHTS

- Delivered 3+ complete end-to-end software projects, including real-time and AI-powered systems
- Improved network uptime by 20% through secure network configuration and proactive monitoring for 30+ staff.
- Resolved hardware, software, and network issues with an average same-day resolution rate of 90%.
- Performed system updates and workstation optimizations, improving device performance by 15%.

WORK EXPERIENCE

Kenya National Highways Authority
Assistant ICT Officer **Jun 2025 - Nov 2025**

- Improved network uptime by 20% through secure network configuration and proactive monitoring for 30+ staff.
- Managed domain accounts and access rights, streamlining user onboarding for 30+ new employees yearly.
- Resolved hardware, software, and network issues with an average same-day resolution rate of 90%.
- Supported digital collaboration by configuring virtual meeting systems used across 5 departments.
- Performed system updates and workstation optimizations, improving device performance by 15%.
- Generated Excel-based operational reports that informed vehicle tracking and resource planning.
- Strengthened system reliability by monitoring ICT assets and reporting incidents promptly.

Kenya National Highways Authority
Administrative Trainee **Apr 2024 - Mar 2025**

- Maintained ICT equipment and network infrastructure, supporting 30+ end users daily.
- Reduced support resolution time by 30% through quick troubleshooting of connectivity and device issues.
- Managed electronic/manual records with 100% data accuracy, improving reporting efficiency.
- Coordinated departmental meetings and logistics, ensuring timely communication and documentation.
- Enhanced workflow efficiency by digitizing 400+ documents and organizing office data systems.
- Supported frontline operations, improving client service turnaround time by 20%.

Kenya National Highways Authority
ICT Intern **Apr 2023 - Mar 2024**

- Assisted in network configuration and troubleshooting, contributing to consistent connectivity uptime of 95%+.
- Supported domain user management for 30+ staff, improving access and account reliability.
- Set up conferencing tools for virtual meetings used across multiple project teams.
- Identified and documented equipment faults, reducing downtime by 25%.
- Performed software installations and updates, boosting workstation efficiency.
- Assisted in ICT inventory tracking for 40+ assets, ensuring accurate documentation.

Kenya National Highways Authority
ICT Attachee **May 2022 - Aug 2022**

- Installed operating systems and essential applications for 10+ workstations, improving user readiness.
- Conducted hardware maintenance and connectivity setup, reducing incident recurrence by 15%.
- Reported faulty components promptly, decreasing overall repair turnaround time.
- Generated vehicle tracking reports supporting operational planning.

- Assisted with ICT asset tracking and documentation, improving inventory accuracy.

EDUCATION

Jomo Kenyatta University of Agriculture and Technology
Bachelor of Business Information Technology
Second Class Honours (Upper Division) GPA **Sep 2019 - Apr 2023**

SKILLS

- React.js
- HTML5
- CSS3
- JavaScript (ES6+)
- TypeScript
- Tailwind CSS
- Node.js
- Express.js
- RESTful API Development
- PostgreSQL
- MySQL
- MongoDB
- Jest
- Postman
- Unit Testing
- Debugging
- Git & Github
- Jenkins
- Docker
- AWS Basics
- Render
- Railway
- Firebase
- Python
- TensorFlow
- API Integration
- UI/UX Collaboration
- Adobe XD
- Agile/Scrum
- Next.js
- Troubleshooting
- End-User Support
- Networking

PROJECTS

Canvas Crafters
https://canvas-crafters.vercel.app/
Designed and built a full-stack, real-time collaborative digital canvas application that enables multiple users to create, edit, and interact on shared canvases simultaneously. Implemented a feature-rich drawing editor with 30+ professional tools, real-time cursor tracking, chat, and permission-based collaboration using WebSockets. Developed a secure backend with user authentication, canvas management, and data persistence, and deployed the system using modern CI/CD workflows. Tech Stack: React, TypeScript, Vite, Tailwind CSS, Node.js, Express, MongoDB, Mongoose, Socket.io, JWT, Vercel, Render, Git

Kilimo AI (Land Degradation Monitoring Platform)
https://v0-remix-of-agri-protect-ai-dashboa.vercel.app/
Designed and built a full-stack AI-powered platform for monitoring land degradation and planning sustainable restoration initiatives. Integrated geospatial mapping, satellite-derived environmental metrics, and machine learning models to analyze soil health, vegetation indices (NDVI), erosion risk, and degradation trends. Implemented AI-driven recommendations to support data-informed land management decisions, and reporting. Tech Stack: Next.js, TypeScript, React, Tailwind CSS, Supabase, Leaflet, Recharts, OpenAI, Vercel AI SDK, Vercel Serverless & Edge Functions, Git

Road Maintenance System
https://road-maintenance-dashboard.netlify.app/
Developed an end-to-end, AI-powered web application for detecting, measuring, and managing road defects from dashcam footage. Implemented a Python-based computer vision pipeline to detect potholes, estimate area, depth, and volume, and classify severity. Built a RESTful backend for data persistence and user management, and designed a responsive dashboard with maps, charts, analytics, and reporting features. Tech Stack: Python, YOLOv8, OpenCV, PyTorch, Node.js, Express, MongoDB, Mongoose, React, TypeScript, Vite, Tailwind CSS, Leaflet, Recharts, REST APIs, Render, Netlify, Git

CERTIFICATIONS

Software Development
Power Learn Project Africa **Feb 2025 - Jul 2025**

AI Literacy Certified Course
Ottermans Institute **Sep 2025 - Oct 2025**

Computer Hardware Certification
CISCO Networking Academy **Aug 2023 - Sep 2023**